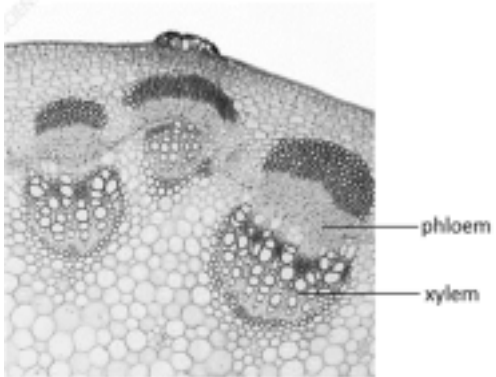


5(a)	phloem (identified) ; xylem (identified) ; e.g. two marks for: 	2
5(b)(i)	any four from: 1 (transport is by) mass flow ; 2 entry of sucrose into phloem sieve tube, from companion cell / through plasmodesmata ; 3 sucrose, is dissolved in water / is part of phloem sap / lowers water potential ; 4 (transport is) down a hydrostatic pressure gradient / from high hydrostatic pressure to low hydrostatic pressure ; <i>high hydrostatic pressure</i> 5 water moves into phloem sieve tube, by osmosis / down water potential gradient / from xylem (at source) ; 6 increased volume increases hydrostatic pressure ; <i>hydrostatic needed once only</i> <i>low hydrostatic pressure</i> 7 (at sink / in other parts of plant) sucrose unloaded, water follows osmotically ; AW	4

Question	Answer	Marks
5(b)(ii)	any two from: <i>respiration</i> 1 less / no, ATP produced from aerobic respiration ; A ETC / oxidative phosphorylation 2 less ATP / energy, for sucrose synthesis (in mesophyll cell) ; <i>in context of companion cell</i> 3 less / no, ATP / energy, for, proton gradient / active transport of protons (out of companion cell) / AW ; A hydrogen ions / H⁺ for protons 4 less sucrose cotransported (with, protons / hydrogen ions / H⁺) into companion cells ; 5 (less sucrose in companion cells means) decreased diffusion of sucrose (via plasmodesmata) into phloem sieve tubes ; <i>if no marks gained, allow 1 mark for less ATP for, active loading / active transport of protons</i>	2
5(c)	Benedict's solution does not react with, non-reducing sugar / sucrose or Benedict's solution only reacts with, reducing sugar / glucose ; AW A Benedict's is, a test for reducing sugars / not a test for non-reducing sugars <i>solution A because</i> sugar present in, cytoplasm / solution A, (mainly) reducing sugars / glucose <u>and</u> sugar in, phloem sap / solution B, is mainly, sucrose / non-reducing sugar or <i>solution B because</i> phloem sap / solution B, contains, a higher concentration of / AW, reducing sugar / glucose, (than, cytoplasm / solution A) ; A in cytoplasm glucose stored as, polysaccharide / starch	2